

1 Books

- (Bretscher, 2009)
- (Covey, 1990)
- (Watts, 1972)
- (Eliot, 1871)
- (Margolis et al., 2011)
- (Duda et al., 2001)
- (Bishop, 2006)
- (Mitchell, 1997)
- (Toffoli & Margolus, 1987)
- (Euclides et al., 2002)
- (Feldman, 2018)
- (Schimmer et al., 2018)
- (Kernighan & Ritchie, 1988)
- (Belvel & Jordan, 2003)
- (Smith, 2004)
- (“Classroom Management That Works”, 2004)
- (Larson et al., 2004b)
- (Larson et al., 2004a)
- (Tucker, 2002)
- (Devlin, 2008)
- (Davis, 2000)
- (Boole, 1847)

2 Papers

- (Wing, 2006)
- (Rabiner, 1989)
- (Gershenson, 2003)
- (de Campos et al., 2009c)

3 Links

- (de Campos et al., 2009a)
- (de Campos et al., 2009b)
- (Bateman & Mayer, 2010)
- (Hubbard & Roby, 2008)
- (Poskanzer, 2009)
- (Goode & Chapman, 2013)
- (Chapman et al., 2025)
- (Center for Technology in Learning at SRI International, 2013)
- (Snow, 2013)
- (Snyder et al., 2013)
- (Seehorn et al., 2011)
- (Stephenson, 2012)

4 Wikipedia

- (Boolean algebra, 2019)
- (Venn diagram, 2019)
- (Complement (set theory), 2019)
- (Naive set theory, 2019)
- (Russell's paradox, 2019)
- (Barber paradox, 2019)
- (Positional notation, 2019)
- (Binary operation, 2019)
- (Unary operation, 2019)
- (Infix notation, 2019)
- (Polish notation, 2019)
- (Bit, 2019)
- (Nybble, 2019)
- (Two's complement, 2019)
- (Ones' complement, 2019)
- (Signedness, 2019)
- (Integer overflow, 2019)
- (Fixed-point arithmetic, 2019)
- (Floating-point arithmetic, 2019)
- (IEEE 754, 2019)
- (Logic gate, 2019)
- (NAND logic, 2019)
- (NOR logic, 2019)
- (Bitwise operation, 2019)
- (Functional completeness, 2019)
- (Adder (electronics), 2019)
- (Multiplication algorithm, 2019)
- (Division algorithm, 2019)
- (Combination, 2010)
- (Combinatorics, 2010)
- (Hungarian algorithm, 2013)
- (Java Programming / Primitive Types, 2018)
- (Permutation, 2010)
- (Plagiarism, 2004)

5 Debug

- (Wikipedia contributors, 2023)
- (Wikipedia contributors, 2005)
- (van de Donk et al., 2004)
- ("This is the title of the web-page", 2009)
- (Jech, 2011)
- (CiteDrive, 2022)

6 MIT App Inventor Research

- (Lee et al., 2026)
- (Zamanova et al., 2026)
- (D. Kim et al., 2026)
- (Afrida et al., 2026)
- (Yuan, 2025)
- (Patton et al., 2025)
- (Ravi et al., 2024)
- (Higman, 2024)
- (Granquist, 2024)
- (Elhashemy, 2024)
- (D. Y. J. Kim et al., 2024)
- (D. Kim et al., 2024)
- (Granquist et al., 2023)
- (Yoo et al., 2023)
- (Iniego, 2023)
- (Elhashemy et al., 2023)
- (Tian, 2023)
- (Pang et al., 2023a)
- (Pang et al., 2023b)
- (Du, Parks, et al., 2023)
- (A. Zhou et al., 2023)
- (Van Brummelen, Tian, et al., 2023)
- (Van Brummelen, Kelleher, et al., 2023)
- (Du, Taylor, et al., 2023)
- (D. Y. J. Kim et al., 2022)
- (Pang et al., 2022)
- (Shone et al., 2022)
- (Adebayo, 2022)
- (Van Brummelen, 2022)
- (Shone, 2022)
- (Pang, 2022)
- (Harkavy, 2022)
- (J. Zhu & Van Brummelen, 2021)
- (T.-C. Hsu, Abelson, & Van Brummelen, 2021)
- (T.-C. Hsu, Abelson, Patton, et al., 2021)
- (J. Zhu, 2021)
- (Yu, 2021)
- (Dunand, 2021)
- (Lin & Van Brummelen, 2021)
- (Van Brummelen, Tabunshchyk, & Heng, 2021)
- (Van Brummelen, Heng, & Tabunshchyk, 2021)
- (T.-C. Hsu, Abelson, Lao, et al., 2021)
- (Lao, 2020)

- (Weng, 2020)
- (Van Brummelen, Weng, et al., 2020)
- (Lin et al., 2020)
- (Van Brummelen, Yeo, & Weng, 2020)
- (Bhatia, 2020)
- (Lao & Bhatia, 2020)
- (Lao et al., 2020)
- (Martin et al., 2020)
- (X. Zhou et al., 2020)
- (Van Brummelen, 2019b)
- (Van Brummelen et al., 2019)
- (Patton, Seo, & Turbak, 2019)
- (Patton, 2019a)
- (Patton, 2019b)
- (D. Tang et al., 2019)
- (Lao et al., 2019)
- (Van Brummelen, 2019a)
- (Tissenbaum et al., 2019)
- (K. Zhu, 2019)
- (D. Tang, 2019)
- (Chan, 2019)
- (Van Brummelen, 2019c)
- (Clarke, 2019)
- (Patton, Tissenbaum, & Harunani, 2019)
- (Tissenbaum et al., 2018)
- (Lao, 2018)
- (Patton, Druga, et al., 2018)
- (Patton, Sherman, & Tissenbaum, 2018)
- (Patton & Tang, 2018)
- (Xi & Patton, 2018)
- (Van Brummelen & Abelson, 2018)
- (Kong et al., 2018)
- (Tseng et al., 2018)
- (L. Tang, 2018)
- (Prutz, 2018)
- (Prutz & Abelson, 2018)
- (Wang, 2018)
- (Sherman & Alkhatib, 2018)
- (Tissenbaum, Sheldon, Seop, et al., 2017)
- (Kong et al., 2017)
- (Ruan et al., 2017)
- (Bau et al., 2017)
- (Tissenbaum, Sheldon, Patton, et al., 2017)
- (Lao, 2017a)
- (Lao, 2017b)
- (Deng, 2017)

- (L. Tang, 2017)
- (Deng & Patton, 2017)
- (Patton & Deng, 2017)
- (Campbell, 2017)
- (Giurleo, 2017)
- (Im, 2017)
- (Li, 2016)
- (Colter, 2016)
- (Xie, 2016)
- (Tsai, 2016)
- (Asmus, 2016)
- (Xie et al., 2015)
- (Morelli et al., 2015)
- (Turbak et al., 2014)
- (Miao, 2014)
- (Wolber et al., 2014)
- (Schiller et al., 2014)
- (Shih & Boortz, 2013)
- (Zhang et al., 2013)
- (Shih et al., 2013)
- (Morelli et al., 2013)
- (Crawford Pokress &
Dominguez Veiga, 2013)
- (Shih & Li, 2013)
- (Y.-C. Hsu & Ching, 2013)
- (Bhagi, 2012)
- (Turbak et al., 2012)
- (Margulieux et al., 2012)
- (Saigal & Saigal, 2011)
- (Magnuson, 2010)
- (Saigal, 2010)

References

- Adder (electronics). (2019, February). In Wikipedia, the free encyclopedia. [https://en.wikipedia.org/w/index.php?title=Adder_\(electronics\)&oldid=879794929](https://en.wikipedia.org/w/index.php?title=Adder_(electronics)&oldid=879794929)
- Adebayo, J. (2022). *Towards effective tools for debugging machine learning models* [Doctoral dissertation, Massachusetts Institute of Technology]. Retrieved September 15, 2026, from https://appinventor.mit.edu/assets/files/AdebayoJulius_PhD_thesis.pdf
- Afrida, A., Patton, E., & Abelson, H. (2026, July 7). AI-driven scaffolding for novice app developers using RAG-finetuned chatbot. In H. Degen & S. Ntoa (Eds.), *Artificial intelligence in HCI* (pp. 375–387, Vol. 16746). Springer, Cham. <https://doi.org/10.1007/978-3-032-31>

048-4_23

Online ISBN is 978-3-032-31048-4 (the print ISBN is in the isbn field above). MIT's research page links the official (paywalled) Springer chapter page instead: https://link.springer.com/chapter/10.1007/978-3-032-31048-4_23; linking the free ResearchGate mirror here.

Asmus, K. (2016). *A system of applications for the integration of BLE beacons in museums* [Master's thesis, Massachusetts Institute of Technology]. Retrieved September 15, 2026, from <https://drive.google.com/open?id=0B3jsksMcCW5bVGZ6WGdJWjUxM1E>

Barber paradox. (2019, February). In Wikipedia, the free encyclopedia. https://en.wikipedia.org/w/index.php?title=Barber_paradox&oldid=883712357

Bateman, W., & Mayer, S. (2010). The online L^AT_EX equation editor. <http://www.codecogs.com/components/equationeditor/equationeditor.php>

Bau, D., Gray, J., Kelleher, C., Sheldon, J., & Turbak, F. (2017). Learnable programming: Blocks and beyond. *Communications of the ACM*, 60(6), 72–80. Retrieved September 15, 2026, from <https://drive.google.com/open?id=0B8AUNoeCnCCAbGVbCUhSNnZDZV>

Belvel, P. S., & Jordan, M. M. (2003). *Rethinking classroom management: Strategies for prevention, intervention, and problem solving*. Corwin Press.

Bhagi, A. (2012). *Android game development with app inventor* [Master's thesis, Massachusetts Institute of Technology]. Retrieved September 15, 2026, from https://appinventor.mit.edu/explore/sites/all/files/Resources/Thesis.FINAL_AnshulBhagi.pdf

Bhatia, N. (2020). *Using transfer learning, spectrogram audio classification, and MIT App Inventor to facilitate machine learning understanding* [Master's thesis, Massachusetts Institute of Technology]. Retrieved September 15, 2026, from https://appinventor.mit.edu/assets/files/Nikhil.Bhatia.MEng_Thesis.pdf

Binary operation. (2019, February). In Wikipedia, the free encyclopedia. https://en.wikipedia.org/w/index.php?title=Binary_operation&oldid=882927455

Bishop, C. M. (2006). *Pattern recognition and machine learning*. Springer Science+Business Media.

Bit. (2019, February). In Wikipedia, the free encyclopedia. <https://en.wikipedia.org/w/index.php?title=Bit&oldid=880961230>

- Bitwise operation. (2019, February). In Wikipedia, the free encyclopedia. https://en.wikipedia.org/w/index.php?title=Bitwise_operation&oldid=882396419
- Boole, G. (1847). *The mathematical analysis of logic: Being an essay towards a calculus of deductive reasoning*. Macmillan, Barclay, & Macmillan.
- Boolean algebra. (2019, February). In Wikipedia, the free encyclopedia. https://en.wikipedia.org/w/index.php?title=Boolean_algebra&oldid=883953881
- Bretscher, O. (2009). *Linear algebra with applications* (4th). Pearson Prentice Hall. <http://isbn.nu/9780136009269>
- Campbell, G. (2017). *Extending cloud data store component for MIT App Inventor* (Term Paper). Massachusetts Institute of Technology. Cambridge, MA. Retrieved September 15, 2026, from <https://drive.google.com/open?id=0B8AUNoeCnCCAbI9iVnJQYnoyaGc>
- Center for Technology in Learning at SRI International. (2013). Draft assessment design patterns, ECS computational thinking practices. http://pact.sri.com/?page_id=832
- Chan, K. K. (2019). *Building an online community of creators through MIT App Inventor* [Master's thesis, Massachusetts Institute of Technology]. Retrieved September 15, 2026, from <https://drive.google.com/open?id=19XiYfawHkFjxDVKgbK9k1hP5VDOjIC5F>
- Chapman, G., Goode, J., & Jayathirtha, G. (2025). Exploring computer science [© University of Oregon, 2025. Labeled a “teacher copy” intended solely for individual classroom use; do not copy or distribute without written permission from ECS.]. Retrieved September 15, 2026, from <https://www.exploringcs.org/wp-content/uploads/2025/09/ECS-v10.pdf>
- CiteDrive, I. (2022). *Citedrive brings reference management to overleaf*. Retrieved August 12, 2023, from <https://www.citedrive.com/overleaf>
- Clarke, N. I. (2019). *Through the screen and into the world: Augmented reality components with MIT App Inventor* [Master's thesis, Massachusetts Institute of Technology]. Retrieved September 15, 2026, from <https://dspace.mit.edu/handle/1721.1/123016>
- Classroom management that works. (2004).
- Colter, A. (2016). *Evaluating and improving the usability of MIT App Inventor* [Master's thesis, Massachusetts Institute of Technology]. Retrieved September 15, 2026, from <https://drive.google.com/open?id=0B3jsksMcCW5bbGw4ZHZvMGQ0bVk>
- Combination. (2010, June). In Wikipedia, the free encyclopedia. <http://en.wikipedia.org/w/index.php?title=Combination&oldid=367895591>

- Combinatorics. (2010, June). In Wikipedia, the free encyclopedia. <http://en.wikipedia.org/w/index.php?title=Combinatorics&oldid=368950754>
- Complement (set theory). (2019, February). In Wikipedia, the free encyclopedia. [https://en.wikipedia.org/w/index.php?title=Complement_\(set_theory\)&oldid=877236947](https://en.wikipedia.org/w/index.php?title=Complement_(set_theory)&oldid=877236947)
- Covey, S. R. (1990). *The 7 habits of highly effective people*. Free Press.
- Crawford Pokress, S., & Dominguez Veiga, J. J. (2013). *MIT App Inventor: Enabling personal mobile computing* [Presented at Programming for Mobile and Touch (PRoMoTo), Indianapolis, IN]. Retrieved September 15, 2026, from <http://arxiv.org/abs/1310.2830>
- Davis, M. (2000). *Engines of logic: Mathematics and the origin of the computer*. W. W. Norton & Company.
- de Campos, T. E., Babu, B. R., & Varma, M. (2009a). The chars74k dataset. <http://www.ee.surrey.ac.uk/CVSSP/demos/chars74k/>
- de Campos, T. E., Babu, B. R., & Varma, M. (2009b). Chars74k hand-drawn characters. <http://www.ee.surrey.ac.uk/CVSSP/demos/chars74k/EnglishHnd.tgz>
- de Campos, T. E., Babu, B. R., & Varma, M. (2009c). Character recognition in natural images. *Proceedings of the International Conference on Computer Vision Theory and Applications, Lisbon, Portugal*.
- Deng, X. (2017). *Group collaboration with app inventor* (Term Paper). Massachusetts Institute of Technology. Cambridge, MA. Retrieved September 15, 2026, from <https://drive.google.com/open?id=0B8AUNoeCnCCANH4X1RkLXh5RTg>
- Deng, X., & Patton, E. W. (2017). Enabling multi-user computational thinking with collaborative blocks programming in MIT App Inventor. *Proceedings of the 1st International Conference on Computational Thinking in Education*. Retrieved September 15, 2026, from https://appinventor.mit.edu/papers/CTE_2017_paper_84.pdf
- Devlin, K. (2008). *The unfinished game: Pascal, Fermat, and the seventeenth-century letter that made the world modern*. Basic Books. <http://isbn.nu/9780465009107>
- Division algorithm. (2019, February). In Wikipedia, the free encyclopedia. https://en.wikipedia.org/w/index.php?title=Division_algorithm&oldid=883467096
- Du, X., Parks, R., Tezel, S., Freilich, J., Pang, H. N., Abelson, H., & Breazeal, C. (2023). Designing a computational action program to tackle global challenges. *Proceedings of the 54th ACM Technical Sym-*

posium on Computer Science Education V. 2, 1320. <https://doi.org/10.1145/3545947.3576267>

- Du, X., Taylor, M., Blumofe, N., Berman, E., Lewallen, N., Geter, S., Park, H. W., Klopfer, E., Abelson, H., & Breazeal, C. (2023). Widening the global access of artificial intelligence (AI) literacy curriculum through the participation of day of AI [Listed as “To Appear”]. *SITE 23 Proceedings*
No online link given on <https://appinventor.mit.edu/explore/research>.
- Duda, R. O., Hart, P. E., & Stork, D. G. (2001). *Pattern classification*. Springer Science+Business Media.
- Dunand, M. (2021). *Developing visual accessibility options to empower grade school students in designing inclusive mobile applications* [Master’s thesis, Massachusetts Institute of Technology]. Retrieved September 15, 2026, from <https://appinventor.mit.edu/assets/files/DunandMEngThesis.pdf>
- Elhashemy, H. (2024). *From curious kids to data wizards: An empowerment approach to data science education* [Master’s thesis, School of Computation, Information and Technology, Technical University of Munich]. Retrieved September 15, 2026, from <https://appinventor.mit.edu/assets/files/ElhashemyMSThesis.pdf>
- Elhashemy, H., Parks, R., Kim, D. Y. J., Patton, E., & Abelson, H. (2023). Empowering learners with a low-barrier mobile data science toolkit. *Constructionism/FabLearn 2023 Proceedings*. Retrieved September 15, 2026, from https://appinventor.mit.edu/assets/files/Elhashemy_FabL-Constr2023.pdf
- Eliot, G. (1871). *Middlemarch*. Penguin Books. <http://isbn.nu/9780141439549>
- Euclides, Densmore, D., & Heath, S. L. (2002). *Euclid’s elements: All thirteen books complete in one volume*. Green Lion Press. <https://books.google.com/books?id=cveqQgAACAAJ>
- Feldman, J. (2018). *Grading for equity: What it is, why it matters, and how it can transform schools and classrooms*. SAGE Publication. <https://isbn.nu/9781506391571>
- Fixed-point arithmetic. (2019, February). In Wikipedia, the free encyclopedia. https://en.wikipedia.org/w/index.php?title=Fixed-point_arithmetic&oldid=874423261
- Floating-point arithmetic. (2019, February). In Wikipedia, the free encyclopedia. https://en.wikipedia.org/w/index.php?title=Floating-point_arithmetic&oldid=883934950

- Functional completeness. (2019, February). In Wikipedia, the free encyclopedia. https://en.wikipedia.org/w/index.php?title=Functional_completeness&oldid=881651954
- Gershenson, C. (2003). Artificial neural networks for beginners. *CoRR*, *cs.NE/0308031*.
- Giurleo, E. (2017). *Keyboard interactions in block-based coding environments* (Term Paper). Massachusetts Institute of Technology. Cambridge, MA. Retrieved September 15, 2026, from <https://drive.google.com/open?id=0B8AUNoeCnCCASzUwMWtwWUhmN1E>
- Goode, J., & Chapman, G. (2013). Exploring computer science. <http://www.exploringcs.org/wp-content/uploads/2010/08/ExploringComputerScience-v5.0.pdf>
- Granquist, A. M. (2024). *AI-augmented interface for incremental app development in MIT App Inventor* [Master’s thesis, Massachusetts Institute of Technology] [Also deposited at <https://dspace.mit.edu/entities/publication/59bb7538-e244-4619-8199-52bf4f150c35>]. Retrieved September 15, 2026, from <https://dspace.mit.edu/handle/1721.1/157025>
- Granquist, A. M., Kim, D. Y. J., & Patton, E. (2023). AI augmented feature to edit and design mobile applications. *MobileHCI 2023 Proceedings*. Retrieved September 15, 2026, from <https://daki7711.github.io/files/granquist2023aiaugmented.pdf>
- Harkavy, E. (2022). *Accessible AI that’s out of this world: Globalizing AI literacy through problem-based learning and deep learning models in a low code environment* [Master’s thesis, Massachusetts Institute of Technology]. Retrieved September 15, 2026, from https://drive.google.com/open?id=1FX0UN4vbbVEJy_uJnc-CZSCpc.udz3hq
MIT M.Eng theses are typically also deposited to DSpace; a separate DSpace copy likely exists but was not independently confirmed.
- Higman, M. (2024). *Designing and writing a JSON AST interpreter for more accessible and extensible IoT communication*. Research Science Institute. Retrieved September 15, 2026, from https://appinventor.mit.edu/assets/files/Higman_Miles_Marissa_Final.pdf
- Hsu, T.-C., Abelson, H., Lao, N., Tseng, Y.-H., & Lin, Y.-T. (2021). Behavioral pattern exploration and development of an instructional tool for young children to learn AI. *Computers and Education: Artificial Intelligence*, 2, 100012. <https://doi.org/10.1016/j.caeai.2021.100012>
- Hsu, T.-C., Abelson, H., Patton, E., Chen, S.-C., & Chang, H.-N. (2021). Self-efficacy and behavior patterns of learners using a real-time collaboration system developed for group programming. *International*

- Journal of Computer-Supported Collaborative Learning*, 16(4), 559–582. <https://doi.org/10.1007/s11412-021-09357-3>
- Hsu, T.-C., Abelson, H., & Van Brummelen, J. (2021). The effects of applying experiential learning into the conversational AI learning platform on secondary school students [Special Issue “AI e-Learning and Online Curriculum”; accepted manuscript / preview version]. *International Review of Research in Open and Distributed Learning (IR-RODL)*. Retrieved September 15, 2026, from https://appinventor.mit.edu/assets/files/JVB_IRRODL_manuscript.docx
- Hsu, Y.-C., & Ching, Y.-H. (2013). Mobile app design for teaching and learning: Educators’ experiences in an online graduate course. *The International Review of Research in Open and Distributed Learning*, 14(4), 117–139. <https://doi.org/10.19173/irrodl.v14i4.1542>
Confirmed against a Zotero-exported BibTeX record; all fields match what was built from the page’s citation metadata.
- Hubbard, M., & Roby, T. (2008). History. <http://binomial.csueastbay.edu/History.html>
- Hungarian algorithm. (2013, May). In Wikipedia, the free encyclopedia. http://en.wikipedia.org/w/index.php?title=Hungarian_algorithm&oldid=553176136
- IEEE 754. (2019, February). In Wikipedia, the free encyclopedia. https://en.wikipedia.org/w/index.php?title=IEEE_754&oldid=883044346
- Im, J. (2017). *User-defined blocks for MIT App Inventor* (Term Paper). Massachusetts Institute of Technology. Cambridge, MA. Retrieved September 15, 2026, from <https://drive.google.com/open?id=0B8AUNoeCnCCAA3lIVkphbkVid3c>
- Infix notation. (2019, February). In Wikipedia, the free encyclopedia. https://en.wikipedia.org/w/index.php?title=Infix_notation&oldid=795311885
- Iniego, A. N. V. (2023). *Leveraging generative AI in college application essay preparation*. Research Science Institute. Retrieved September 15, 2026, from https://appinventor.mit.edu/assets/files/Iniego_Angel_Nicole_Shuvom_Final.pdf
- Integer overflow. (2019, February). In Wikipedia, the free encyclopedia. https://en.wikipedia.org/w/index.php?title=Integer_overflow&oldid=880564879
- Java Programming / Primitive Types. (2018, November). In Wikibooks, the free textbook project. https://en.wikibooks.org/w/index.php?title=Java_Programming/Primitive_Types&oldid=3491137

- Jech, T. (2011). Set theory. In E. N. Zalta (Ed.), *The stanford encyclopedia of philosophy* (Winter 2011). Retrieved April 23, 2014, from <http://plato.stanford.edu/archives/win2011/entries/set-theory/>
- Kernighan, B. W., & Ritchie, D. M. (1988). *The C programming language* (Second). Prentice-Hall, Inc.
- Kim, D., Bagiati, A., Mishra, T., Alghowinem, S., Taylor, M., Burghleh, T., Freilich, J., Tezel, S., Parks, R., & Breazeal, C. (2026–October 14). Futuremakers: An innovative approach to online AI education [Accepted paper.]. *Proceedings of IEEE Frontiers in Education 2026 (FIE 2026)*
No online link given on <https://appinventor.mit.edu/explore/research>.
- Kim, D., Zhou, A., Sudo, Y., & Takano, K. (2024). Advancing mobile app development and generative AI education through MIT App Inventor. *CTE-STEM 2024 Proceedings*. Retrieved September 15, 2026, from <https://appinventor.mit.edu/assets/files/kim2024advancing.pdf>
- Kim, D. Y. J., Granquist, A. M., Patton, E., Friedman, M., & Abelson, H. (2022). Speak your mind: Introducing APTLY, the software platform that turns ideas into working apps. *iCERi 2022 Proceedings*. Retrieved September 15, 2026, from <https://daki7711.github.io/files/ICERIpaper2022.pdf>
- Kim, D. Y. J., Ravi, P., Williams, R., & Yoo, D. (2024). App planner: Utilizing generative AI in K-12 mobile app development education. *Proceedings of the 23rd Annual ACM Interaction Design and Children Conference*, 770–775. <https://doi.org/10.1145/3628516.3659392>
- Kong, S. C., Abelson, H., Sheldon, J., Lao, A., Tissenbaum, M., & Lai, M. (2017). Curriculum activities to foster primary school students’ computational practices in block-based programming environments. *International Conference on Computational Thinking Education 2017 (CTE 2017)*, 84
No online link given on <https://appinventor.mit.edu/explore/research>.
- Kong, S. C., Andone, D., Biswas, G., Crick, T., Hoppe, H. U., Hsu, T. C., Huang, R. H., Li, K. Y., Looi, C. K., Milrad, M., Sheldon, J., Shih, J. L., Sin, K. F., Tissenbaum, M., & Vahrenhold, J. (2018). On tools that support the development of computational thinking skills: Some thoughts and future vision [Editors’ foreword to the CTE2018 proceedings volume]. In S. C. Kong, D. Andone, G. Biswas, T. Crick, H. U. Hoppe, T. C. Hsu, R. H. Huang, K. Y. Li, C. K. Looi, M. Milrad, J. Sheldon, J. L. Shih, K. F. Sin, M. Tissenbaum, & J.

- Vahrenhold (Eds.), *Proceedings of the international conference on computational thinking education 2018*. The Education University of Hong Kong. Retrieved September 15, 2026, from https://drive.google.com/open?id=18MMtXBfghN6_nlAl47pCCrxQHwCsLXfG
- Lao, N. (2017a). *CloudDB: Components for exploring shared data with MIT App Inventor* [Describes ongoing development of the CloudDB component.]. Retrieved September 15, 2026, from https://drive.google.com/open?id=1pJa_0gbFCWazcFVBfJXHGbTyjdjuJmXA
No formal venue given on <https://appinventor.mit.edu/explore/research>.
- Lao, N. (2017b). *Developing cloud and shared data capabilities to support primary school students in creating mobile applications that affect their communities* [Master's thesis, Massachusetts Institute of Technology]. Retrieved September 15, 2026, from <https://drive.google.com/open?id=0B8AUNoeCnCCAVDZxV2I3NS1qbHc>
- Lao, N. (2018). CloudDB in MIT App Inventor [Demo abstract.]. *Proceedings of the BLOCKS+ Workshop, co-located with SPLASH 2018*
No online link given on <https://appinventor.mit.edu/explore/research>.
- Lao, N. (2020). *Reorienting machine learning education towards tinkerers and ML-engaged citizens* [Doctoral dissertation, Massachusetts Institute of Technology]. Retrieved September 15, 2026, from https://appinventor.mit.edu/assets/files/NatalieLao_PhD_Dissertation.pdf
- Lao, N., & Bhatia, N. (2020). Using transfer learning, spectrogram audio classification, and MIT App Inventor to facilitate machine learning understanding. *International Conference on Computational Thinking Education 2020 (CTE2020)*. Retrieved September 15, 2026, from https://appinventor.mit.edu/assets/files/CTE2020_paper_38.pdf
- Lao, N., Lee, I., & Abelson, H. (2019). A deep learning practicum: Concepts and practices for teaching actionable machine learning at the tertiary education level. *ICERI 2019 Conference*. Retrieved September 15, 2026, from https://drive.google.com/open?id=13N_uS1wFbxbMzER0D8GPZf-ksB8eKomp
- Lao, N., Lee, I., & Abelson, H. (2020). Experiences from teaching actionable machine learning at the university level through a small practicum approach. *International Conference on Computational Thinking Education 2020 (CTE2020)*. Retrieved September 15, 2026, from https://appinventor.mit.edu/assets/files/CTE2020_paper_9.pdf
- Larson, R., Boswell, L., & Stiff, L. (2004a). *Geometry*. McDougal Littell.

- Larson, R., Boswell, L., & Stiff, L. (2004b). *Geometry: Concepts and skills*. Houghton Mifflin School.
- Lee, V. R., Madaio, M., Garside, B., Welch, A., Blair, K. P., Adisa, I. O., Harris, A., Holst, K., Rafael, L. B., Morad, R. L., Travis, B., Moller, B., Shields, A., Brown, Z., Hinx, L., Diaz, M., Patton, E., Tezel, S., Parks, R., . . . Roschelle, J. (2026–June 19). Google, AI literacy, and the learning sciences: Multiple modes of research, industry, and practice partnerships. *International Society of the Learning Sciences (ISLS 2026)*. Retrieved September 15, 2026, from <https://arxiv.org/abs/2604.07601>
- Li, W. (2016). *Helping the helpers: A toolkit for mobile humanitarian assistance apps* [Master’s thesis, Massachusetts Institute of Technology]. Retrieved September 15, 2026, from <https://drive.google.com/open?id=0B3jsksMcCW5bMFdKLWh2T1I3LWc>
- Lin, P., & Van Brummelen, J. (2021). Engaging teachers to co-design integrated AI curriculum for K-12 classrooms. *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*, 239:1–239:12. Retrieved September 15, 2026, from <https://arxiv.org/abs/2009.11100>
- Lin, P., Van Brummelen, J., Lukin, G., Williams, R., & Breazeal, C. (2020). Zhorai: Designing a conversational agent for children to explore machine learning concepts. *Tenth Symposium on Educational Advances in Artificial Intelligence (EAAI)*. Retrieved September 15, 2026, from https://appinventor.mit.edu/papers/JessVBPublications/zhorai_conversational_agent_ML.pdf
- Logic gate. (2019, February). In Wikipedia, the free encyclopedia. https://en.wikipedia.org/w/index.php?title=Logic_gate&oldid=883992939
- Magnuson, B. (2010). *Building blocks for mobile games: A multiplayer framework for app inventor for android* [Master’s thesis, Massachusetts Institute of Technology]. Retrieved September 15, 2026, from <http://dspace.mit.edu/handle/1721.1/61253>
- Margolis, J., Estrella, R., Goode, J., Holme, J. J., & Nao, K. (2011). *Stuck in the shallow end*. MIT Press. <http://isbn.nu/978-0-262-51404-0>
- Margulieux, L. E., Guzdial, M., & Catrambone, R. (2012). Subgoal-labeled instructional material improves performance and transfer in learning to develop mobile applications. *Proceedings of the Ninth Annual International Conference on International Computing Education Research*, 71–78. <https://doi.org/10.1145/2361276.2361291>
- Martin, F., Lee, I., Lytle, N., Sentance, S., & Lao, N. (2020). Panel: Extending and evaluating the use-modify-create progression for engaging

- youth in computational thinking. *SIGCSE '20*. Retrieved September 15, 2026, from <https://appinventor.mit.edu/assets/files/SIGCSE2020Panel.pdf>
- Miao, D. Y. (2014). *PrivacyInformer: An automated privacy description generator for the MIT App Inventor* [Master's thesis, Massachusetts Institute of Technology]. Retrieved September 15, 2026, from <http://dig.csail.mit.edu/2014/Theses/ThesisDanielaMiao.pdf>
- Mitchell, T. M. (1997). *Machine learning*. McGraw-Hill Companies, Inc.
- Morelli, R., Uche, C., Lake, P., & Baldwin, L. (2015). Analyzing year one of a CS principles project. *46th ACM Technical Symposium on Computer Science Education*, 368–373. Retrieved September 15, 2026, from <https://drive.google.com/open?id=0B3jsksMcCW5bTFBWSEZyNU5DRkk>
- Morelli, R., Wolber, D., Pokress, S., Turbak, F., & Martin, F. (2013). Teaching the CS principles curriculum with app inventor (abstract only). *Proceedings of the 44th ACM Technical Symposium on Computer Science Education*, 762. <https://doi.org/10.1145/2445196.2445516>
- Multiplication algorithm. (2019, February). In Wikipedia, the free encyclopedia. https://en.wikipedia.org/w/index.php?title=Multiplication_algorithm&oldid=875028575
- Naive set theory. (2019, February). In Wikipedia, the free encyclopedia. https://en.wikipedia.org/w/index.php?title=Naive_set_theory&oldid=878461198
- NAND logic. (2019, February). In Wikipedia, the free encyclopedia. https://en.wikipedia.org/w/index.php?title=NAND_logic&oldid=874024831
- NOR logic. (2019, February). In Wikipedia, the free encyclopedia. https://en.wikipedia.org/w/index.php?title=NOR_logic&oldid=882243128
- Nybble. (2019, February). In Wikipedia, the free encyclopedia. <https://en.wikipedia.org/w/index.php?title=Nybble&oldid=883088049>
- Ones' complement. (2019, February). In Wikipedia, the free encyclopedia. https://en.wikipedia.org/w/index.php?title=Ones'_complement&oldid=872645202
- Pang, H. N. (2022). *Computational action in action: Process and tools that empower students to make a real-world impact using technology* [Master's thesis, Massachusetts Institute of Technology]. Retrieved September 15, 2026, from <https://dspace.mit.edu/handle/1721.1/144881>
- Pang, H. N., Parks, R., Breazeal, C., & Abelson, H. (2022). The effect of computational action on students' software prototype ideas. *ICERI2022 Proceedings*, 1804–1814. <https://doi.org/10.21125/iceri.2022.0460>

MIT’s research page lists this paper under its 2023 heading, though it was presented at ICERI 2022, per IATED’s own record.

- Pang, H. N., Parks, R., Breazeal, C., & Abelson, H. (2023a). The effect of computational action on students’ computational identity and self-efficacy. *EDULEARN23 Proceedings*, 8404–8412. <https://doi.org/10.21125/edulearn.2023.2183>
- Pang, H. N., Parks, R., Breazeal, C., & Abelson, H. (2023b). “How Can I Code A.I. Responsibly?”: The effect of computational action on K-12 students learning and creating socially responsible A.I. *EAAI 23 Proceedings*. Retrieved September 15, 2026, from <https://ojs.aaai.org/index.php/AAAI/article/view/26902>
- Patton, E. W. (2019a). Exploring blockly performance: Case studies in MIT App Inventor. *Blockly User Summit 2019*. Retrieved September 15, 2026, from <https://drive.google.com/open?id=1JmZ93iQOZ5LWj8kjU9AfwVEND39bwT0v>
MIT’s research page links the same file via the older <https://drive.google.com/file/d/1JmZ93iQOZ5LWj8kjU9AfwVEND39bwT0v/view> form.
- Patton, E. W. (2019b). Realtime collaboration with blockly: Experiences with MIT App Inventor. *Blockly User Summit 2019*. Retrieved September 15, 2026, from https://drive.google.com/open?id=1Yq9_Ay4a0paFgpVQR1pJN5_wzfUtEaBW
MIT’s research page links the same file via the older https://drive.google.com/file/d/1Yq9_Ay4a0paFgpVQR1pJN5_wzfUtEaBW/view form.
- Patton, E. W., & Deng, X. (2017). Collaborative mobile app development with MIT App Inventor [Demo presentation.]. *Proceedings of Blocks & Beyond 2017*
No online link given on <https://appinventor.mit.edu/explore/research>.
- Patton, E. W., Druga, S., Van Brummelen, J., Mazzone, J., & Puig Fernandez, X. (2018). Panel: Block abstractions for artificial intelligence. *Proceedings of the BLOCKS+ Workshop, co-located with SPLASH 2018*. Retrieved September 15, 2026, from https://appinventor.mit.edu/papers/blocksplus18_p14_patton.pdf
- Patton, E. W., Kim, D. Y. J., Granquist, A. M., Liu, R., Scott, A., Zamanova, J., & Abelson, H. (2025). Aptly: Making mobile apps from natural language. *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA ’25)*. <https://doi.org/10.1145/3706599.3720081>

- Patton, E. W., Seo, A., & Turbak, F. (2019). Enhancing abstraction in app inventor with generic event handlers. *Blocks & Beyond 2019*. Retrieved September 15, 2026, from <https://appinventor.mit.edu/papers/s18-patton.pdf>
- Patton, E. W., Sherman, M., & Tissenbaum, M. (2018). Research tools for MIT App Inventor. *Proceedings of the BLOCKS+ Workshop, co-located with SPLASH 2018*. Retrieved September 15, 2026, from https://appinventor.mit.edu/papers/blocksplus18_p15-patton.pdf
- Patton, E. W., & Tang, D. (2018). JSON interoperability in MIT App Inventor. *Proceedings of the BLOCKS+ Workshop, co-located with SPLASH 2018*. Retrieved September 15, 2026, from https://appinventor.mit.edu/papers/blocksplus18_p13-tang.pdf
- Patton, E. W., Tissenbaum, M., & Harunani, F. (2019). MIT App Inventor: Objectives, design, and development. In S. C. Kong & H. Abelson (Eds.), *Computational thinking education* (pp. 31–49). Springer Singapore. https://doi.org/10.1007/978-981-13-6528-7_3
- Permutation. (2010, June). In Wikipedia, the free encyclopedia. <http://en.wikipedia.org/w/index.php?title=Permutation&oldid=366563313>
- Plagiarism. (2004, June). In Wikipedia, the free encyclopedia. <http://en.wikipedia.org/w/index.php?title=Plagiarism&oldid=5139350>
- Polish notation. (2019, February). In Wikipedia, the free encyclopedia. https://en.wikipedia.org/w/index.php?title=Polish_notation&oldid=880582128
- Positional notation. (2019, February). In Wikipedia, the free encyclopedia. https://en.wikipedia.org/w/index.php?title=Positional_notation&oldid=877761722
- Poskanzer, J. (2009). Acme license maker. <http://www.acme.com/licensemaker/>
- Prutz, K. (2018). *Democratizing cross-platform mobile development with MIT App Inventor* (Term Paper). Massachusetts Institute of Technology. Cambridge, MA. Retrieved September 15, 2026, from https://drive.google.com/open?id=1xCwT3fpXdAupVI5oE5_-dcmOaw0z149H
MIT’s research page dates this “Spring 2018” with no specific month given.
- Prutz, K., & Abelson, H. (2018, May 17). *Expanding device functionality for the MIT App Inventor IoT embedded companion* (Term Paper). Massachusetts Institute of Technology. Cambridge, MA. Retrieved September 15, 2026, from <https://drive.google.com/open?id=1AV1Za0bnoIHioj9SS0unIkU0aKT3DPBg>

- Rabiner, L. R. (1989). A tutorial on hidden markov models and selected applications in speech recognition. *Proceedings of the IEEE*. http://www.cs.cornell.edu/Courses/cs4758/2012sp/materials/hmm_paper_rabiner.pdf
- Ravi, P., Parks, R., Masla, J., Abelson, H., & Breazeal, C. (2024). “Data Comes from the Real World”: A constructionist approach to mainstreaming K-12 data science education. *SIGCSE Virtual 2024: Proceedings of the 2024 on ACM Virtual Global Computing Education Conference*. Retrieved September 15, 2026, from https://appinventor.mit.edu/assets/files/SIGCSE_2024_Data_Science.pdf
- Ruan, L. L., Patton, E. W., & Tissenbaum, M. (2017). Evaluations of programming complexity in app inventor. *International Conference on Computational Thinking Education 2017 (CTE 2017)*. Retrieved September 15, 2026, from <https://drive.google.com/open?id=0B8AUNoeCnCCARUMtS0NqN3R5d1E>
- Russell’s paradox. (2019, February). In Wikipedia, the free encyclopedia. https://en.wikipedia.org/w/index.php?title=Russell’s_paradox&oldid=872268102
- Saigal, A. K. (2010). *Information accountability for mobile financial applications* [Master’s thesis, Massachusetts Institute of Technology]. Retrieved September 15, 2026, from <http://dspace.mit.edu/handle/1721.1/85493>
Listed under the “2013” heading on MIT’s research page, but the thesis text itself states “Cambridge, 2010”; filed here by actual date.
- Saigal, A. K., & Saigal, A. (2011). Saathimobile and the rapid deployment of prototypes to build applications for social enterprise in the developing world. *Proceedings of the 1st International Conference on Wireless Technologies for Humanitarian Relief*, 351–356. <https://doi.org/10.1145/2185216.2185311>
- Schiller, J., Turbak, F., Friedman, M., Abelson, H., Okerlund, J., Dominguez, J., & McKinney, A. (2014). Live programming of mobile apps in app inventor. *Programming for Mobile and Touch (PROMOTO)*, 1–8. Retrieved September 15, 2026, from <https://drive.google.com/open?id=0B3jsksMcCW5bS2k3TXpKVIFYX28>
The old <https://docs.google.com/file/d/0B3jsksMcCW5bS2k3TXpKVIFYX28> form on MIT’s research page could not be independently confirmed still resolving; the drive.google.com/open?id=0B3jsksMcCW5bS2k3TXpKVIFYX28 form above has been confirmed working.
- Schimmer, T., Hillman, G., & Stalets, M. (2018). *Standards-based learning in action: Moving from theory to practice (a guide to implementing*

- standards-based grading, instruction, and learning*). Solution Tree. <https://isbn.nu/9781945349010>
- Seehorn, D., Carey, S., Fuschetto, B., Moix, I. L. D., O’Grady-Cunniff, D., Owens, B. B., Stephenson, C., & Verno, A. (2011). CSTA k-12 computer science standards. <http://csta.acm.org/Curriculum/sub/K12Standards.html>
- Sherman, M., & Alkhatib, O. M. (2018, August 17). *MIT analysis of the 2018 technovation student projects, sponsored by salesforce* (“Assisted by Obada M. Alkhatib”). MIT App Inventor. Retrieved September 15, 2026, from <https://appinventor.mit.edu/assets/files/MIT-Salesforce-Technovation-Report-2018-1-1.pdf>
- Shih, F., & Boortz, J. (2013). Understanding people’s preferences for disclosing contextual information to smartphone apps. In L. Marinos & I. Askoxylakis (Eds.), *Human aspects of information security, privacy, and trust* (pp. 186–196). Springer Berlin Heidelberg. https://doi.org/10.1007/978-3-642-39345-7_20
- Shih, F., & Li, W. (2013). Unlocking smartphone data for educational use in teaching and learning environment. *Smart University Workshop, Eighth International and Interdisciplinary Conference on Modeling and Using Context*. Retrieved September 15, 2026, from <https://appinventor.mit.edu/explore/sites/explore.appinventor.mit.edu/files/resources/smartUniv.pdf>
- Shih, F., Seneviratne, O., Miao, D., Liccardi, I., Kagal, L., Patton, E. W., Meier, P., & Castillo, C. (2013). Democratizing mobile app development for disaster management. *AIIP Workshop on Artificial Intelligence Problems and Approaches for Intelligent Environments and Workshop on Semantic Cities*, 39–42. Retrieved September 15, 2026, from <http://dig.csail.mit.edu/2013/QCRI-DIG-project/paper.pdf>
- Shone, J. (2022). *Yes, you can make an app too: A systematic study of prompt engineering in the automatic generation of mobile applications from user queries*. Research Science Institute. Retrieved September 15, 2026, from https://appinventor.mit.edu/assets/files/Shone_Jasmine_Rachel_RSIFinal.pdf
- Shone, J., Liu, R., Patton, E., & Kim, D. Y. J. (2022). Exploring prompt engineering for generative AI-based app generation. *Acta Scientific Computer Sciences*, 5(5). Retrieved September 15, 2026, from <http://actascientific.com/ASCS/ASCS-05-0448.php>
- Signedness. (2019, February). In Wikipedia, the free encyclopedia. <https://en.wikipedia.org/w/index.php?title=Signedness&oldid=792830866>

- Smith, R. (2004). *Conscious classroom management: Unlocking the secrets of great teaching*. Conscious Teaching Publications.
- Snow, D. (2013, November). PACT presentation materials from CE21 2013. http://pact.sri.com/?page_id=1285
- Snyder, L., Astrachan, O., & Briggs, A. (2013). Seven big ideas of computer science. <http://csprinciples.cs.washington.edu/sevenbigideas.html>
- Stephenson, C. (2012, November). The national imperative for k–12 computer science education. http://www.csta.acm.org/Advocacy_Outreach/sub/PresentationFiles/MicrosoftPanel_Stephenson.pdf
- Tang, D. (2019). *Empowering novices to understand and use machine learning with personalized image classification models, intuitive analysis tools, and MIT App Inventor* [Master’s thesis, Massachusetts Institute of Technology]. Retrieved September 15, 2026, from <https://drive.google.com/open?id=1fOVkajhYZaFtVGZnNWjQn6nBrBLWzE1>
- Tang, D., Utsumi, Y., & Lao, N. (2019). PIC: A personal image classification webtool for high school students. *IJCAI 2019 EduAI Workshop*. Retrieved September 15, 2026, from https://drive.google.com/open?id=18x1pGEoKrQ_4ShkFdH2hYjoTtc0qyECa
- Tang, L. (2017). *Creating customized user interfaces for app inventor* (Term Paper). Massachusetts Institute of Technology. Cambridge, MA. Retrieved September 15, 2026, from <https://drive.google.com/open?id=0B8AUNoeCnCCAN2ZaeFRHcENWbGM>
- Tang, L. (2018). *Constrained sets: The effects of multi-layered environments in learning app inventor* [Advisor: Harold Abelson.]. Retrieved September 15, 2026, from <https://dspace.mit.edu/handle/1721.1/119707>
- Exact degree level and defense date not confirmed.
- This is the title of the webpage.* (2009, February). Retrieved August 29, 2013, from <http://www.example.com>
- Tian, M. C. (2023). *The design and implementation of stateful ConvoBlocks* [BA thesis]. Wellesley College. Retrieved September 15, 2026, from https://appinventor.mit.edu/assets/files/WCTC_2023-TianMingyan_TheDesignAndImp.pdf
- Tissenbaum, M., Sheldon, J., & Abelson, H. (2018). Reducing the barriers for computational action: How blocks-based programming can support digital empowerment. *Proceedings of the BLOCKS+ Workshop, co-located with SPLASH 2018*. Retrieved September 15, 2026, from <https://appinventor.mit.edu/papers/blocksplus18-paper18.pdf>

- Tissenbaum, M., Sheldon, J., & Abelson, H. (2019). From computational thinking to computational action. *Communications of the ACM*, 62(3), 34–36. <https://doi.org/10.1145/3265747>
- Tissenbaum, M., Sheldon, J., Patton, E. W., Gupta, A., Zhang, E., & Gopinath, D. (2017). Off the screen, and into the world of everyday objects: Computational thinking for youth with the internet of things. *International Conference on Computational Thinking Education 2017 (CTE 2017)*. Retrieved September 15, 2026, from <https://drive.google.com/open?id=0B8AUNoeCnCCAOUx2NFU5UzExY1E>
- Tissenbaum, M., Sheldon, J., Seop, L., Lee, C., & Lao, N. (2017). Critical computational empowerment: Engaging youth as shapers of the digital future. *Global Engineering Education Conference (EDUCON), 2017 IEEE*, 1705–1708. <https://doi.org/10.1109/EDUCON.2017.7943078>
IEEE Xplore (<https://ieeexplore.ieee.org/document/7943078/>) is paywalled; linking the free full-text mirror instead.
- Toffoli, T., & Margolus, N. (1987). *Cellular automata machines: A new environment for modeling*. The MIT Press.
- Tsai, S. (2016). *Understanding app inventor forums* [Master’s thesis, Massachusetts Institute of Technology]. Retrieved September 15, 2026, from <https://drive.google.com/open?id=0B3jsksMcCW5bS1Z4VkJmaGRCY1E>
- Tseng, C. H., Tissenbaum, M., Kuan, W. H., Hsu, F. C., & Wong, C. C. (2018). A design-based approach to implementing a computational thinking curriculum with app inventor and the internet of things. *International Conference on Computational Thinking Education 2018 (CTE 2018)*. Retrieved September 15, 2026, from <https://drive.google.com/open?id=1P-jl1gNdkFIxExEeMdo9qiNcvdaLJbxnl>
- Tucker, A. (2002). *Applied combinatorics* (Fourth). John Wiley & Sons. <http://isbn.nu/047143809X>
- Turbak, F., Sandu, S., Kotsopoulos, O., Erdman, E., Davis, E., & Chadha, K. (2012). Blocks languages for creating tangible artifacts. *IEEE Symposium on Visual Languages and Human-Centric Computing*, 137–144. Retrieved September 15, 2026, from <http://cs.wellesley.edu/~fturbak/pubs/VLHCC-2012-paper-turbak.pdf>
- Turbak, F., Sherman, M., Martin, F., Wolber, D., & Crawford Pokress, S. (2014). Events-first programming in app inventor. *Journal of Computing Sciences in Colleges*, 29(6), 81–89. Retrieved September 15,

- 2026, from <https://drive.google.com/open?id=0B3jsksMcCW5bSGhUUjctMkFXT1E>
- Two’s complement. (2019, February). In Wikipedia, the free encyclopedia. https://en.wikipedia.org/w/index.php?title=Two's_complement&oldid=882476598
- Unary operation. (2019, February). In Wikipedia, the free encyclopedia. https://en.wikipedia.org/w/index.php?title=Unary_operation&oldid=876888704
- Van Brummelen, J., & Abelson, H. (2018). “What’s Conversational AI?” with MIT App Inventor and amazon alexa [Poster]. *Amazon Research Days 2018*. Retrieved September 15, 2026, from https://appinventor.mit.edu/explore/sites/explore.appinventor.mit.edu/files/Amazon_Alexa_Day_Poster_portrait.pdf
- Van Brummelen, J. (2019a). Conversational agents to democratize artificial intelligence. *IEEE Symposium on Visual Languages and Human-Centric Computing (VL/HCC)*, 239–240. <https://doi.org/10.1109/VLHCC.2019.8818805>
- Van Brummelen, J. (2019b). Conversational artificial intelligence development tools for K-12 education. *AAAI Fall Symposium*. Retrieved September 15, 2026, from https://appinventor.mit.edu/papers/JessVBPublications/AAAI_Fall_Symposium_2019_SMConvAI_Final.pdf
- Van Brummelen, J. (2019c). *Tools to create and democratize conversational artificial intelligence* [Master’s thesis, Massachusetts Institute of Technology]. Retrieved September 15, 2026, from <https://drive.google.com/open?id=1AMNP3hvfS1EEgnwFugJkIjEpQ3KMtFQk>
Explicitly labeled an M.S. thesis, distinct from the M.Eng degree used for most theses in this bibliography, and keyed accordingly (**ms** rather than **meng**). MIT M.S./M.Eng theses are typically also deposited to DSpace; a separate DSpace copy likely exists but was not independently confirmed.
- Van Brummelen, J. (2022). *Empowering K-12 students to understand and design conversational agents: Concepts, recommendations and development platforms* [Doctoral dissertation, Massachusetts Institute of Technology]. Retrieved September 15, 2026, from <https://drive.google.com/open?id=16SkT81IKeiz0Hmu6euoNzvJEbDso5XME>
MIT PhD dissertations are typically also deposited to DSpace; a separate DSpace copy likely exists but was not independently confirmed.
- Van Brummelen, J., Heng, T., & Tabunshchik, V. (2021). Teaching tech to talk: K-12 conversational artificial intelligence literacy curriculum

and development tools. *Proceedings of the AAAI Conference on Artificial Intelligence*, 35(17), 15655–15663. <https://doi.org/10.1609/aaai.v35i17.17844>

MIT’s research page still links only the <https://arxiv.org/pdf/2009.05653.pdf> preprint; this points to the confirmed published AAAI version instead.

- Van Brummelen, J., Kelleher, M., Tian, M. C., & Nguyen, N. H. (2023). What do children and parents want and perceive in conversational agents? towards transparent, trustworthy, democratized agents. *Proceedings of the 22nd Annual ACM Interaction Design and Children Conference*, 187–197. <https://doi.org/10.1145/3585088.3589353>
- Van Brummelen, J., Shen, J. H., & Patton, E. W. (2019). The popstar, the poet, and the grinch: Relating artificial intelligence to the computational thinking framework with block-based coding. *International Conference on Computational Thinking Education 2019 (CTE2019)*. Retrieved September 15, 2026, from https://appinventor.mit.edu/papers/JessVBPublications/Popstar_Poet_Grinch_CTE2019.pdf
- Van Brummelen, J., Tabunshchik, V., & Heng, T. (2021). “Alexa, Can I Program You?”: Student perceptions of conversational artificial intelligence before and after programming alexa. *Proceedings of the 20th Annual ACM Interaction Design and Children Conference*, 305–313. <https://doi.org/10.1145/3459990.3460730>
- Van Brummelen, J., Tian, M. C., Kelleher, M., & Nguyen, N. H. (2023). Learning affects trust: Design recommendations and concepts for teaching children — and nearly anyone — about conversational agents. *Proceedings of the AAAI Conference on Artificial Intelligence*, 37, 15860–15868. Retrieved September 15, 2026, from <https://arxiv.org/pdf/2209.05063.pdf>
- Van Brummelen, J., Weng, K., Lin, P., & Yeo, C. (2020). CONVO: What does conversational programming need? *IEEE Symposium on Visual Languages and Human-Centric Computing (VL/HCC)*, 1–5. <https://doi.org/10.1109/VL/HCC50065.2020.9127277>
- Van Brummelen, J., Yeo, C., & Weng, K. (2020). Learning to program conversationally: A conversational agent to further democratize programming. *14th International Technology, Education and Development Conference (INTED2020)*. Retrieved September 15, 2026, from https://appinventor.mit.edu/papers/JessVBPublications/INTED2020_abstract.pdf

- van de Donk, W., Loader, B. D., Nixon, P. G., & Rucht, D. (Eds.). (2004). *Cyberprotest: New media, citizens and social movements*. Routledge. Retrieved October 29, 2013, from www.jstor.com/somearticle.html
- Venn diagram. (2019, February). In Wikipedia, the free encyclopedia. https://en.wikipedia.org/w/index.php?title=Venn_diagram&oldid=881835160
- Wang, K. (2018). *Creating a database log history page for cloud components in app inventor to support software development education* (Term Paper). Massachusetts Institute of Technology. Cambridge, MA. Retrieved September 15, 2026, from <https://drive.google.com/open?id=0BzGL9U0OLqkQUdxQ281Z3B6U1A4VE42UzVNZ1hGb0pKUkx3>
- Watts, A. (1972). *The supreme identity*. Vintage Books.
- Weng, K. (2020). *Examining conversational programming design needs with convo, a voice-first conversational programming system using natural language* [Master's thesis, Massachusetts Institute of Technology]. Retrieved September 15, 2026, from https://appinventor.mit.edu/assets/files/MEng.Thesis_Kevin.Weng.pdf
- Wikipedia contributors. (2005, August). *Wikipedia* [[Online; accessed 14-August-2023]]. Retrieved August 14, 2023, from <https://en.wikipedia.org/Wikipedia>
- Wikipedia contributors. (2023). Carmichael number — Wikipedia, the free encyclopedia [[Online; accessed 13-August-2023]].
- Wing, J. M. (2006). Computational thinking. *Communications of the ACM*, 49(3).
- Wolber, D., Abelson, H., & Friedman, M. (2014). Democratizing computing with app inventor. *GetMobile: Mobile Computing and Communications*, 18(4), 53–58. Retrieved September 15, 2026, from <https://drive.google.com/open?id=0B3jsksMcCW5baTlwSEpVWjBlQ00>
The old <https://docs.google.com/file/d/0B3jsksMcCW5baTlwSEpVWjBlQ00> form on MIT's research page could not be independently confirmed still resolving; the drive.google.com/open?id=0B3jsksMcCW5baTlwSEpVWjBlQ00 form above has been confirmed working.
- Xi, W., & Patton, E. W. (2018). A blocks-based approach to internet of things in MIT App Inventor. *Proceedings of the BLOCKS+ Workshop, co-located with SPLASH 2018*. Retrieved September 15, 2026, from https://appinventor.mit.edu/papers/blocksplus18_p12_xi.pdf
- Xie, B. (2016). *Progression of computational thinking skills demonstrated by app inventor users* [Master's thesis, Massachusetts Institute of

- Technology]. Retrieved September 15, 2026, from <https://drive.google.com/open?id=0B3jsksMcCW5bZGVuemF5VDQ0QWc>
- Xie, B., Shabir, I., & Abelson, H. (2015). Measuring the usability and capability of app inventor to create mobile applications. *PROMOTO 2015*, 108. Retrieved September 15, 2026, from <http://dspace.mit.edu/handle/1721.1/98913>
- Yoo, D., Kim, D. Y. J., & Lopes, E. (2023). Digital art therapy with gen AI: Mind palette. *11th International Conference on Affective Computing and Intelligent Interaction Workshops and Demos (ACIIW)*. Retrieved September 15, 2026, from <https://appinventor.mit.edu/assets/files/yoo2023digital.pdf>
- Yu, J. (2021). *Using facemesh in MIT App Inventor to empower students to apply artificial intelligence* [Master’s thesis, Massachusetts Institute of Technology]. Retrieved September 15, 2026, from <https://drive.google.com/open?id=1-QBYzYf5uReYSCT6f5OOI2VPDijXggE-> MIT’s research page links the same file via the older <https://drive.google.com/file/d/1-QBYzYf5uReYSCT6f5OOI2VPDijXggE-/view?pli=1> form.
- Yuan, J. (2025, May). *Empowering mobile-only app generation — Offline AI code generation with App Inventor* [Master’s thesis, Massachusetts Institute of Technology]. Retrieved September 15, 2026, from <http://appinventor.mit.edu/assets/files/Joyce2025Thesis.pdf> Thesis supervisor: Harold Abelson, Class of 1922 Professor of Computer Science and Engineering. Accepted by: Katrina LaCurts, Chair, Master of Engineering Thesis Committee.
- Zamanova, J., Patton, E. W., & Abelson, H. (2026). *Visual project comparison system for block-based graphical programming platforms* [Submitted.]. No online link given on <https://appinventor.mit.edu/explore/research>.
- Zhang, F., Shih, F., & Weitzner, D. (2013). No surprises: Measuring intrusiveness of smartphone applications by detecting objective context deviations. *Proceedings of the 12th ACM Workshop on Workshop on Privacy in the Electronic Society*, 291–296. <https://doi.org/10.1145/2517840.2517864>
- Zhou, A., Kim, D. Y. J., & Abelson, H. (2023). Learning and using image classifiers by creating mobile applications. *INTED2023 Proceedings*. Retrieved September 15, 2026, from https://docs.google.com/document/d/1f0LKoJKpiF--_DlnXU0pIEEnANthFSjyj-Z1MNkte3m4/edit

- Zhou, X., Van Brummelen, J., & Lin, P. (2020). *Designing AI learning experiences for K-12: Emerging works, future opportunities and a design framework* [Unpublished preprint; no subsequent peer-reviewed venue found as of 2026]. Retrieved September 15, 2026, from <https://arxiv.org/pdf/2009.10228.pdf>
- Zhu, J. (2021). *Creating your own conversational artificial intelligence agents using convo, a conversational programming system* [Master’s thesis, Massachusetts Institute of Technology]. Retrieved September 15, 2026, from https://appinventor.mit.edu/assets/files/JessicaZhu_MEng.Thesis.pdf
- Zhu, J., & Van Brummelen, J. (2021). Teaching students about conversational AI using convo, a conversational programming agent. *IEEE Symposium on Visual Languages and Human-Centric Computing (VL/HCC)*, 1–5. <https://doi.org/10.1109/VL/HCC51201.2021.9576290>
- Zhu, K. (2019). *An educational approach to machine learning with mobile applications* [Master’s thesis, Massachusetts Institute of Technology]. Retrieved September 15, 2026, from <https://drive.google.com/open?id=1Q0D2wyiPLfpZzFc0cxNk5jl9PWuN3oYl>